Rui Zhu

Applied Scientist II, Amazon
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Research Interests

DNA large language models; Security, safety and privacy of large language models.

Education

2019 – 2024 **Ph.D., Indiana University Bloomington**, Department of Computer Science

2017 – 2019 **M.S., Indiana University Bloomington**, Department of Statistics

2011 – 2015 **B.S., University of Utah**, Department of Mathematics

Work Experience

2026 Mar – Now **Applied Scientist II** in Amazon;
Medical Foundation Model; Built and trained a medical foundation model from scratch with multiple modules integrating heterogeneous data sources, including EHR records, claims data, drug SMILES, and patient genomic data. Processed massive datasets and optimized system performance for high-throughput workloads, adhering to stringent privacy and compliance requirements.

Affiliated Lecturer in Yale School of Medicine;

2024 Jun – 2026 Feb **Post-doctoral Researcher** in Yale University, School of Medicine, *Advisor: Dr. Lucila Ohno-Machado*;
DNA Large Language Model & Privacy; Designed downstream applications based on a DNA large language model (Evo2) with a strong focus on genomic privacy protection: performed continued pretraining on human pan-genome data to "humanize" the model while mitigating privacy risks in human genetic variation. Conducted supervised fine-tuning and rigorous model evaluation for secure, privacy-preserving clinical (highly regulated) prediction tasks.
LLM Model Compression; Implemented model compression for the DNA LLM (Evo2) to enable more efficient inference on much longer genomic sequences, optimizing system performance and scaling solutions for high-stakes contexts.
Tabular Large Foundation Model; Built applications on top of the tabular foundation model TabPFN by fine-tuning it on clinical datasets, achieving performance that outperforms standard gradient-boosting–based methods.

2023 May-August **Software Engineer** in The Trade Desk
Adversarial Model; Developed a deep learning–based adversarial model to predict competitors (such as Google Ads) bidding for on-page impressions. Designed secure APIs, integrated the ML model with back-end systems, and deployed robust, scalable AI-driven solutions to production.

2018 May-August **Data Scientist** Union Pay Lab
Community detection; To build this community detection model. One patent was applied in this project(Patent number: CN09493086A, Patent page: http://www.soopat.com/Patent/201811261870).

Work Experience (continued)

2016 – 2017 **Data Scientist** in Lenovo;
Text Classification; Text classification model for online sentiment analysis of customer comments in e-commerce.
Fake Comments Detection in E-commerce; Identify whether a comment from e-commerce is from a real user or a machine.
Retailer Fraud Detection; To detect Insurance and warranty frauds in Retailers of Lenovo.

Teaching Experience

Lecturer **I599: Data Science on-ramp** – Fall 2019 - Fall 2020 (3 semesters)
I529: Machine Learning in Bioinformatics – Fall 2023 - Spring 2024

Achievements

Apr 2023 **Indiana University Bloomignton**: Best Research Assistant of the year 2022 in computer science department

Patent

community detection model

2018.11 Patent number: CN09493086A, Patent page: http://www.soopat.com/Patent/201811261870

Projects

TrojAI project by NIST and IARPA (DoD/DARPA equivalent)

2020.5 – 2024.5 In team Perspectra-IUB, Main contributor. Main weekly meeting reporter. Got 2 out of 16 teams in round 3 competition and 3 out of 16 teams in round 10 competition. Evaluated and assessed AI model vulnerabilities (Trojans/Backdoors). Developed robust evaluation frameworks to ensure AI safety and security under stringent government and DoD standards.

Publications

** indicates co-first authors.*

Selected Publications

1. Wang*, Z., **Zhu***, R., Zhang, Z., Tang, H., & Wang, X. (2025). Rigging the foundation: Manipulating pre-training for advanced membership inference attacks. *IEEE S&P*.
2. Wang*, Z., **Zhu***, R., Zhou, D., Zhang, Z., Wang, X., & Tang, H. (2025). Sharpness-aware initialization: Improving differentially private machine learning from first principles. *34th USENIX Security Symposium (USENIX Security 2025)*, 3103–3122.
3. Chen*, X., Tang*, S., **Zhu***, R., Yan, S., Jin, L., Wang, Z., Su, L., Zhang, Z., Wang, X., & Tang, H. (2024). The janus interface: How fine-tuning in large language models amplifies the privacy risks. *Proceedings of the 2024 ACM SIGSAC Conference on Computer and Communications Security (CCS)*, 1285–1299.
4. **Zhu**, R., Tang, D., Tang, S., Wang, Z., Tao, G., Ma, S., Wang, X., & Tang, H. (2024). Gradient shaping: Enhancing backdoor attack against reverse engineering. *NDSS*.

5 Wang*, Z., **Zhu***, R., Zhou, D., Zhang, Z., Mitchell, J., Tang, H., & Wang, X. (2024). Dpadapter: Improving differentially private deep learning through noise tolerance pre-training. *USENIX Security*.

6 **Zhu***, R., Tang*, D., Tang, S., Wang, X., & Tang, H. (2023). Selective amnesia: On efficient, high-fidelity and blind suppression of backdoor effects in trojaned machine learning models. *IEEE S&P*.

Trustworthy AI and Data

1 Fan*, Y., **Zhu***, R., Wang, Z., Wang, C., Tang, H., Dong, Y., Cho, H., & Ohno-Machado, L. (2025). Byzsfl: Achieving byzantine-robust secure federated learning with zero-knowledge proofs. *arXiv preprint arXiv:2501.06953*.

2 Li, A., Qian, L., Du, M., Yin, Y., Hu, Y., **Zhu**, R., Sun, Z., Fu, Y., Stutz, E., Ai, X., Xie, Q., et al. (2025). Memorization in large language models in medicine: Prevalence, characteristics, and implications. *arXiv preprint arXiv:2509.08604 (Nature Communication Accepted)*.

3 Li, X., Song, K., **Zhu**, R., Chen, P.-Y., & Tang, H. (2025). Adversarial attack-defense co-evolution for llm safety alignment via tree-group dual-aware search and optimization. *arXiv preprint arXiv:2511.19218*.

4 **Zhu**, R., Mortensen, G., Fan, Y., & Tang, H. (2025). Illuminating the unseen: A large-scale exploration of bias in icu discharge summaries via language models. *IEEE BHI*.

5 Wang*, Z., **Zhu***, R., Zhang, Z., Tang, H., & Wang, X. (2025). Rigging the foundation: Manipulating pre-training for advanced membership inference attacks. *IEEE S&P*.

6 Wang*, Z., **Zhu***, R., Zhou, D., Zhang, Z., Wang, X., & Tang, H. (2025). Sharpness-aware initialization: Improving differentially private machine learning from first principles. *34th USENIX Security Symposium (USENIX Security 2025)*, 3103–3122.

7 Xiong*, C., Wang*, Z., **Zhu***, R., Ho, T.-Y., Chen, P.-Y., Xiong, J., Tang, H., & Ohno-Machado, L. (2025). Hey, that's my data! label-only dataset inference in large language models. *arXiv preprint arXiv:2506.06057 (ACL 2026 Accepted)*.

8 Chen*, X., Tang*, S., **Zhu***, R., Yan, S., Jin, L., Wang, Z., Su, L., Zhang, Z., Wang, X., & Tang, H. (2024). The janus interface: How fine-tuning in large language models amplifies the privacy risks. *Proceedings of the 2024 ACM SIGSAC Conference on Computer and Communications Security (CCS)*, 1285–1299.

9 Tang, D., **Zhu**, R., Chen, Y., Tang, H., & Wang, X. (2024). Understanding impacts of task similarity on backdoor attack and detection. *arXiv preprint arXiv:2210.06509*.

10 **Zhu**, R., Tang, D., Tang, S., Wang, Z., Tao, G., Ma, S., Wang, X., & Tang, H. (2024). Gradient shaping: Enhancing backdoor attack against reverse engineering. *NDSS*.

11 Wang*, Z., **Zhu***, R., Zhou, D., Zhang, Z., Mitchell, J., Tang, H., & Wang, X. (2024). Dpadapter: Improving differentially private deep learning through noise tolerance pre-training. *USENIX Security*.

12 Li*, Z., **Zhu***, R., Wang, Z., Tang, H., & Chai, H. (2023). Fairfix: Improving fairness of deep neural networks through selective amnesia and recovery. *SmartCom (Best Paper Award)*.

13 **Zhu***, R., Tang*, D., Tang, S., Wang, X., & Tang, H. (2023). Selective amnesia: On efficient, high-fidelity and blind suppression of backdoor effects in trojaned machine learning models. *IEEE S&P*.

14 **Zhu**, R., Jiang, C., Wang, X., Wang, S., Zheng, H., & Tang, H. (2020). Privacy-preserving construction of generalized linear mixed model for biomedical computation. *Bioinformatics, ISMB2020, 36*(Suppl 1), i128.

AI for Science

1 **Zhu**, R., Zhou, X., Liang, I., Stephen, S., Xu, K., & Ohno-Machado, L. (2026a). Foundation model–guided synthetic ehr release: Performance enhancement with privacy preservation. *AMIA Amplify Informatics Summit*.

2. **Zhu**, R., Zhou, X., Liang, I., Stephen, S., Xu, K., & Ohno-Machado, L. (2026b). Large models for small tables: Adapting tabular foundation models to ehr data. *AMIA Amplify Informatics Summit*.
3. Ge, C., Xiong, J., **Zhu**, R., Hong, Z., & He, Y. (2025a). The global burden of high bmi among adolescents between 1990 and 2021. *Communications Medicine*, *5*(1), 125.
4. Ge, C., Xiong, J., **Zhu**, R., Hong, Z., & He, Y. (2025b). Global burden of high sugar-sweetened beverage consumption among young adults. *Diabetology & Metabolic Syndrome*, *17*(1), 259.
5. Ma*, X., **Zhu, R.***, Wang, Z., Xiong, J., Chen, Q., Tang, H., Camp, L. J., & Ohno-Machado, L. (2025). Enhancing patient-centric communication: Leveraging llms to simulate patient perspectives. *arXiv preprint arXiv:2501.06964*.
6. Mortensen*, G., & **Zhu***, R. (2025). Adetectolocum: Harnessing locally deployable llms for early alzheimer's disease detection through voice analysis. *AMIA Informatics Summit 2025*.
7. **Zhu***, R., Zhou*, X., Aquino, M. M. d., Engchuan, W., Zhou, H., Cheng, H., Initiative, A. D. N., Hardy, J., Tang, H., Scherer, S. W., et al. (2025). Advancing human population genomics with dna foundation models. *medRxiv (Nature Communication under review)*, 2025–09.
8. **Zhu**, R., Zhou, X., Tang, H., Scherer, S. W., & Ohno-Machado, L. (2025). Near-lossless model compression enables longer context inference in dna large language models. *arXiv preprint arXiv:2511.14694 (RECOMB 2026 under review)*.
9. Zhang, K., **Zhu**, R., Ma, S., Xiong, J., Kim, Y., Murai, F., & Liu, X. (2025). Kedrec-lm: A knowledge-distilled explainable drug recommendation large language model. *NIPIR Workshop*.

Others

1. Li, X., Jiao, W., Song, K., **Zhu**, R., & Tang, H. (2025). Knowledge-aware co-reasoning for multidisciplinary collaboration. *EMNLP*.
2. Li, X., Song, K., **Zhu**, R., Tang, H., et al. (2025). Knowledge-aware co-reasoning for multidisciplinary collaboration. *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing*, 13615–13631.
3. Li, X., Liu, K., **Zhu**, R., & Liu, X. (2023). Hierarchical multi-task learning for enterprise risk detection from financial documents. *IEEE BigData*.
4. Li, X., Qin, Y., **Zhu**, R., Lin, T., Fan, Y., Kang, Y., Song, K., Zhao, F., Sun, C., Tang, H., & Liu, X. (2023). Stinmatch: Semi-supervised semantic-topological iteration network for financial risk detection via news label diffusion. *EMNLP*.
5. **Zhu**, R., Lin, B., & Tang, H. (2020). Bounding the number of linear regions in local area for neural networks with relu activations. *arXiv preprint arXiv:2007.06803*.

Media Coverage

2023

- **The New York Times**: Interviewed by NYT journalist Jeremy White and featured in the front-page story "How strangers got my email address from ChatGPT's model" `https://www.nytimes.com/interactive/2023/12/22/technology/openai-chatgpt-privacy-exploit.html`
- **Business Today**: "Email used to log in for ChatGPT at potential risk, research team unveils OpenAI's GPT-3.5 Turbo threat" `https://www.businesstoday.in/technology/news/story/email-used-to-log-in-for-chatgpt-at-potential-risk-research-team-unveils-openais-gpt-35-turbo-threat-410774-2023-12-26`

Service To Professional Community

- **Served as Subject Matter Expert**: Patient-Centered Outcomes Research Institute (PCORI)
- **Served as Associate Editor**: IEEE Engineering in Medicine and Biology Society (IEEE EMBC)
- **Served as Chair**: NDSS Worshop on AI System with Confidential Computing (AISCC 2024)
- **Served as Organizer**: integrating Data for Analysis, Anonymization and Sharing (iDASH 2024)
- **Served as PC**: International Conference on Database Systems for Advanced Applications (DASFAA 2024)
- **Served as PC**: ACM Conference on Data and Application Security and Privacy (CODASPY)
- **Served as Reviewer**: Bioinformatics Advances
- **Served as Reviewer**: IEEE Transactions on Information Forensics & Security
- **Served as PC**: ACM Conference on Data and Application Security and Privacy (CODASPY)
- **Served as Reviewer**:IEEE Journal of Biomedical and Health Informatics
- **Served as Artifact PC member**: Annual Computer Security Applications Conference (ACSAC)
- **Served as Organizer and PC**:Backdoor Attacks aNd DefenSes in Machine Learning (BANDS) Workshop at ICLR 2023
- **Served as Reviewer**:Intelligent Systems for Molecular Biology
- **Served as PC member**: Bioinformatics journal
- **Served as Artifact PC member**: Annual Computer Security Applications Conference (ACSAC)
- **Served as PC member and session Chair**: European Symposium on Research in Computer Security
- **Served as PC member and session Chair**: European Symposium on Research in Computer Security